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GCP ACE Cert prep sheet

Mentor contact info:
Name: Sanjeev Singh
Email: sssskumar@google.com | sanjeev@gcptrain.org |
https://www.linkedin.com/in/sanjeevvepsun/

Partner support team- cloud-partner-training@google.com

Thank you for attending the workshop. I would appreciate any feedback you could share, as your perspective is valuable for my continuous improvement.
https://partner.skills.google/classrooms/32935/course_surveys/cloud-training-l2io16cn5i

To create sample/mock exam

https://gemini.google.com/app

"create practice exam for gcp ACE cert exam "

GCP Certs Exam vouchers for free -

https://cloud.google.com/resources/pck-page?hl=en

No Recording as per NDA

Here's a form that learners can fill out to request access to Qwiklabs class
https://docs.google.com/forms/d/e/1FAIpQLSfW2qgpU9sSgwZI_yy-gtAiWlo0L_BU-GT7squnpZMQzUOhuA/viewform
Training code - PCA25522
Trainer Name - Sanjeev

Colour code Meaning
Green - Recommended for ACE certification by Mentor
Yellow- Additional Learning by Mentor
Red- Must for Certification Voucher/Coupon by Google

****

Self paced learning (labs)

https://partner.skills.google/paths/69

Note: Use your **partner email address** (**work email address**) to login to the portal to download the slides and also to access the self-paced learning & skillbadges.

ACE new exam Guide/Syllabus

https://services.google.com/fh/files/misc/associate_cloud_engineer_exam_guide_english.pdf

https://cloud.google.com/learn/certification/cloud-engineer

Sample/Practice Questions & Answers

https://docs.google.com/forms/d/e/1FAIpQLSfexWKtXT2OSFJ-obA4iT3GmzgiOCGvjrT9OfxilWC1yPtmfQ/viewform **** (19 QA)

https://partner.cloudskillsboost.google/course_templates/1055 ********
(350 + QA)

Create your own mock exam using below prompt in Gemini

https://gemini.google.com/app

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GenAI learning workshops
https://rsvp.withgoogle.com/events/partner-learning/gen-ai-schedule

Additional certifications:

https://rsvp.withgoogle.com/events/partner-learning/home

Certification
https://rsvp.withgoogle.com/events/partner-learning/partner-certification-academy

Solutioning
https://rsvp.withgoogle.com/events/partner-learning/solution-training-schedule

Google Product
https://rsvp.withgoogle.com/events/partner-learning/product-training-schedule

Basic Networking(Homework)

https://www.youtube.com/watch?v=cNb7xKyya5c&list=PLIivdWyY5sqJ0oXcnZYqOnuNRsLF9H48u&index=3

Networking in Google Cloud: Fundamentals

https://partner.cloudskillsboost.google/paths/79/course_templates/35

Networking in Google Cloud: Routing and Addressing

https://partner.cloudskillsboost.google/paths/79/course_templates/36

Networking in Google Cloud: Network Architecture

https://partner.cloudskillsboost.google/paths/79/course_templates/1144

GCP Cheatsheet

https://cloud.google.com/static/sdk/docs/images/gcloud-cheat-sheet.pdf

https://github.com/priyankavergadia/GCPSketchnote

https://www.thecloudgirl.dev/

https://www.thecloudgirl.dev/videos

https://drive.google.com/file/d/1YVxQOgYR5xMVE6A6DSWh-GcjzK4TQ4vp/view?usp=sharing

https://cloud.google.com/products/?hl=en

Databases

https://cloud.google.com/blog/topics/developers-practitioners/your-google-cloud-database-options-explained

https://cloud.google.com/products/databases?hl=en

Cloud SQL Auth Proxy: https://cloud.google.com/sql/docs/mysql/sql-proxy

Private services access :https://cloud.google.com/vpc/docs/private-services-access

Private Google Access: https://cloud.google.com/vpc/docs/private-google-access

Private Service Connect: https://cloud.google.com/vpc/docs/private-service-connect

Security Guide

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https://www.youtube.com/watch?v=X-kZ_eNtxyU&list=WL&index=5

https://www.youtube.com/watch?v=xXk1YlkKW_k&list=WL&index=4

https://partner.cloudskillsboost.google/course_templates/702?catalog_rank=%7B%22rank%22%3A15%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=

https://cloud.google.com/binary-authorization/docs/reference/organizational-and-iam-roles

https://cloud.google.com/iam/docs/using-iam-securely#least_privilege

https://cloud.google.com/iam/docs/service-account-overview

https://www.youtube.com/watch?v=X-kZ_eNtxyU&list=WL&index=6

https://www.youtube.com/playlist?list=PLIivdWyY5sqIIx9XF5lKbs5pr3PEn0QLI

Certmetrics

https://cp.certmetrics.com/google/en/login

OS login

https://cloud.google.com/compute/docs/oslogin?_gl=1*16f3rko*_ga*MzE1OTk2MzE5LjE3MzMxOTU0MTQ.*_ga_WH2QY8WWF5*MTczMzMwMDY3Mi4zLjEuMTczMzMwM

GKE vs Cloud Rub vs Function vs App Engine

https://tutorialsdojo.com/google-cloud-functions-vs-app-engine-vs-cloud-run-vs-gke/

****

GKE Optional learning

Manage scalable workload on GKE (3 days training -24 hrs)
https://rsvp.withgoogle.com/events/partner-learning/solution-training-schedule

Architecting with GKE (2 days -16hrs training)
https://rsvp.withgoogle.com/events/partner-learning/product-training-schedule

Docker labs *****

https://partner.cloudskillsboost.google/focuses/11621?catalog_rank=%7B%22rank%22%3A1%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=

https://partner.cloudskillsboost.google/focuses/60091?catalog_rank=%7B%22rank%22%3A2%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=

Cheatsheet CMD Cli
https://docs.docker.com/get-started/docker_cheatsheet.pdf
https://www.bluematador.com/learn/kubectl-cheatsheet
https://cloud.google.com/static/sdk/docs/images/gcloud-cheat-sheet.pdf

https://cloud.google.com/static/sdk/docs/images/gcloud-cheat-sheet.pdf

GKE cmd Cheatsheet : https://www.bluematador.com/learn/kubectl-cheatsheet

https://www.youtube.com/watch?v=XsH7mKXD_y4&list=PLIivdWyY5sqLQ3m7WJDfBdMMqO12Q0vqg

https://www.youtube.com/watch?v=znhnDHAPCZE&list=PLibPu7aGZPlfolTcU0oZRcFwd48JLQYUm

https://partner.cloudskillsboost.google/course_templates/34?catalog_rank=%7B%22rank%22%3A5%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_

https://partner.cloudskillsboost.google/focuses/11641?catalog_rank=%7B%22rank%22%3A1%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=

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https://partner.cloudskillsboost.google/course_templates/660

https://partner.cloudskillsboost.google/focuses/73819?catalog_rank=%7B%22rank%22%3A3%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=cata

https://cloud.google.com/kubernetes-engine/docs/concepts/service

https://partner.cloudskillsboost.google/focuses/101678?catalog_rank=%7B%22rank%22%3A1%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=cata

https://cloud.google.com/learn/certification?hl=en

https://rsvp.withgoogle.com/events/partner-learning/product-training-schedule

https://partner.cloudskillsboost.google/course_templates/744?catalog_rank=%7B%22rank%22%3A2%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495058

Aap on GKE
https://partner.cloudskillsboost.google/course_templates/663?catalog_rank=%7B%22rank%22%3A3%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495058

https://partner.cloudskillsboost.google/course_templates/655?catalog_rank=%7B%22rank%22%3A5%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495058

https://partner.cloudskillsboost.google/course_templates/655?catalog_rank=%7B%22rank%22%3A5%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495058

https://partner.cloudskillsboost.google/course_templates/2?catalog_rank=%7B%22rank%22%3A46%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495076

https://partner.cloudskillsboost.google/course_templates/783?catalog_rank=%7B%22rank%22%3A47%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495076

https://partner.cloudskillsboost.google/course_templates/34?catalog_rank=%7B%22rank%22%3A48%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495076

https://partner.cloudskillsboost.google/course_templates/32?catalog_rank=%7B%22rank%22%3A49%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495076

https://partner.cloudskillsboost.google/course_templates/33?catalog_rank=%7B%22rank%22%3A52%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495085

https://partner.cloudskillsboost.google/course_templates/660?catalog_rank=%7B%22rank%22%3A56%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495085

https://partner.cloudskillsboost.google/course_templates/951?catalog_rank=%7B%22rank%22%3A60%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=38495085

https://partner.cloudskillsboost.google/course_templates/725?catalog_rank=%7B%22rank%22%3A12%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&search_id=

****

Optional

https://partner.cloudskillsboost.google/course_templates/1145

https://partner.cloudskillsboost.google/course_templates/502

https://www.virtasant.com/blog/stateful-vs-stateless-architecture-why-stateless-won

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Keeping Informed:

- Google Cloud Console: https://console.cloud.google.com/
- **GCP blog: https://cloud.google.com/blog/
- Google's developer blog: https://developers.googleblog.com/
- What's new in GCP: https://cloud.google.com/blog/topics/inside-google-cloud/whats-new-google-cloud
- Developer Discord channel: https://discord.com/invite/google-dev-community
- GCP community support: https://cloud.google.com/support/docs/community-support
- Digital transformation with Google Cloud: https://cloud.google.com/transform/
- Google AI blog: https://ai.googleblog.com/
- GCP newsletter: https://cloud.google.com/newsletter
- GCP YouTube channel: https://www.youtube.com/user/googlecloudplatform
- GCP Medium page: https://medium.com/google-cloud
- Recent GCP product changes (all release notes): https://cloud.google.com/release-notes
- **Product specific release notes: https://cloud.google.com/release-notes/all
- Google Cloud Podcasts: https://cloud.google.com/podcasts
- **Security bulletins: https://cloud.google.com/support/bulletins
- GCP on Reddit: https://www.reddit.com/r/googlecloud/

Starter information:

- **Google Cloud Architecture Center: https://cloud.google.com/architecture
  - What's new in the Architecture center: https://cloud.google.com/architecture/release-notes
- **Google Cloud products: https://cloud.google.com/products
- **Google Regions: https://cloud.google.com/about/locations
  - us-east1 (Moncks CorGlobal Locations - Regions & Zones | Google Cloudner, SC)
  - us-central1 (Council Bluffs, IA)
  - europe-west1 (Saint Ghislain, Belgium)
- **Which resources are zonal, regional, multi-regional, or global: https://cloud.google.com/compute/docs/regions-zones/global-regional-zonal-resources
- **Relate Azure, AWS, and GCP service names: https://cloud.google.com/free/docs/aws-azure-gcp-service-comparison
- **Google cloud developer's cheat sheet (GCP product list):
  - https://github.com/priyankavergadia/google-cloud-4-words
  - Products: https://cloud.google.com/products/?hl=en
- **Great graphics related to Google Cloud: https://thecloudgirl.dev/
  - https://github.com/priyankavergadia/GCPSketchnote
- GCP Solutions: https://cloud.google.com/solutions
- GCP Tutorials: https://cloud.google.com/docs/tutorials
- Google cloud design icons: https://cloud.google.com/icons
- General Googly (material) icons: https://fonts.google.com/icons
- Architecture diagram tool: https://googlecloudcheatsheet.withgoogle.com/architecture
- About Google Cloud Services: https://cloud.google.com/docs/overview/cloud-platform-services
- Architecture framework: https://cloud.google.com/architecture/framework
- **Check latency from you to every GCP Region:

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- Google datacenter photo gallery: https://www.google.com/about/datacenters/gallery/
- GCP Certification: https://cloud.google.com/certification
- Google's "green" status: https://cloud.google.com/sustainability/region-carbon
- Case studies and customers: https://cloud.google.com/customers/
- Find a Google partner: https://cloud.google.com/find-a-partner/
- Google Cloud Support hub: https://cloud.google.com/support-hub
- Customer care packages: https://cloud.google.com/support/docs
- Google Cloud Issue tracker: https://cloud.google.com/support/docs/issue-trackers
- Wild and crazy cloud computing stats: https://www.cloudzero.com/blog/cloud-computing-statistics
- Nice page of best practices links: https://ronnieroyston.com/google-cloud/google-cloud-best-practices

Cloud Shell / SDK:

- Installing the GCP Cloud SDK: https://cloud.google.com/sdk/install
- gcloud cheat sheet:
  - https://cloud.google.com/sdk/docs/cheatsheet
  - https://cloud.google.com/sdk/docs/images/gcloud-cheat-sheet.pdf
  - https://gist.github.com/pydevops/cffbd3c694d599c6ca18342d3625af97
- gcloud reference: https://cloud.google.com/sdk/gcloud/reference
- gsutil docs: https://cloud.google.com/storage/docs/gsutil
  - Consider the new gcloud storage: https://cloud.google.com/sdk/gcloud/reference/storage
- bq command line reference: https://cloud.google.com/bigquery/docs/reference/bq-cli-reference
- Interactive GCP cloud SDK: https://cloud.google.com/sdk/docs/interactive-gcloud
- How Cloud Shell works: https://cloud.google.com/shell/docs/how-cloud-shell-works

Compute Technologies:

- **Nice "where should I run my stuff" sketch: https://www.thecloudgirl.dev/compute/where-should-i-run-my-stuff
- **Compute tech overview: https://cloud.google.com/docs/compute-area
- Compute related sketches: https://www.thecloudgirl.dev/compute

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- Standard: https://cloud.google.com/appengine/docs/standard
- Supported Runtimes: https://cloud.google.com/appengine/docs/standard/runtimes
- Flex: https://cloud.google.com/appengine/docs/flexible

Cloud Functions:

- Cloud Functions: https://cloud.google.com/functions/docs/concepts/overview
- Supported environments: https://cloud.google.com/functions/docs/concepts/exec
- Deploying functions: https://cloud.google.com/functions/docs/deploy
- Tips and tricks on using: https://cloud.google.com/functions/docs/bestpractices/tips

Compute Engine:

- General info: https://cloud.google.com/compute
- GCE Overview: https://cloud.google.com/compute/docs/overview
- Compute Engine instances overview: https://cloud.google.com/compute/docs/instances
- GCE Instance Groups: https://cloud.google.com/compute/docs/instance-groups
    - Instance templates: https://cloud.google.com/compute/docs/instance-templates/create-instance-templates
    - Autoscaling options: https://cloud.google.com/compute/docs/autoscaler
- GCE Machine Types: https://cloud.google.com/compute/docs/machine-resource
- CPU platform details: https://cloud.google.com/compute/docs/cpu-platforms
- GCE Storage (drive) types: https://cloud.google.com/compute/docs/disks
- What is persistent disk (Cloud Girl): https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/pd.jpg
- Persistent disk performance: https://cloud.google.com/compute/docs/disks/performance
- Image management best practices: https://cloud.google.com/compute/docs/images/image-management-best-practices
- Migrate for Compute Engine: https://cloud.google.com/migrate/compute-engine/docs/
- OS Login (leverage IAM to manage SSH access:
    - https://cloud.google.com/compute/docs/instances/access-overview
    - https://cloud.google.com/compute/docs/oslogin
- Shielded VMs for better bootkit and rootkit protection: https://cloud.google.com/compute/shielded-vm/docs/shielded-

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- Snapshots: https://cloud.google.com/compute/docs/disks/create-snapshots
- Placement policies: https://cloud.google.com/compute/docs/instances/define-instance-placement
- Zone virtualization: https://cloud.google.com/compute/docs/regions-zones/zone-virtualization
- Regional machine type comparison: https://gcloud-compute.com/

Cloud Run:

- Cloud Run: https://cloud.google.com/run/docs/overview/what-is-cloud-run
- Deploy an app to Cloud Run with Cloud Deploy: https://cloud.google.com/deploy/docs/deploy-app-run
- Cloud Run service identity: https://cloud.google.com/run/docs/securing/service-identity
- Sample code: https://cloud.google.com/run/docs/samples
- Secrets: https://cloud.google.com/run/docs/configuring/services/secrets
- Managed Knative: https://knative.dev/
    - Knative serving (Cloud Run Anthos): https://cloud.google.com/anthos/run/docs/install
- CloudEvents: https://cloudevents.io/
- Eventarc (Cloud Run triggers):https://cloud.google.com/eventarc/docs/overview
- Cloud Run ingress options: https://cloud.google.com/run/docs/securing/ingress
- Cloud Run and Cloud Armor: https://cloud.google.com/armor/docs/integrating-cloud-armor#serverless
- Execution environment options: https://cloud.google.com/run/docs/about-execution-environments
- VPC Connection options: https://cloud.google.com/run/docs/configuring/connecting-vpc
- How to talk to which services: https://cloud.google.com/run/docs/integrate/using-gcp-services
- Private Networking and Cloud Run: https://cloud.google.com/run/docs/securing/private-networking

Kubernetes:

Google's Kubernetes Podcast: https://kubernetespodcast.com/

- What is a container: https://www.docker.com/resources/what-container
- Open Container Initiative (OCI): https://opencontainers.org/
- Containerd graphic: https://containerd.io/img/architecture.png
- The differences between Docker, containerd, CRI-O and runc: https://www.tutorialworks.com/difference-docker-containerd-

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type=image

- Dockerfile reference: https://docs.docker.com/engine/reference/builder/
- Docker commands cheat sheet:
  - https://dockerlabs.collabnix.com/docker/cheatsheet/
  - https://docs.docker.com/get-started/docker_cheatsheet.pdf
- Dockerfile best practices:
  - https://docs.docker.com/develop/develop-images/dockerfile_best-practices/
  - https://dev.to/chrisedrego/21-best-practise-in-2021-for-dockerfile-1dji
- Kubernetes is deprecating Docker (Don't panic!): https://kubernetes.io/blog/2020/12/02/dont-panic-kubernetes-and-docker/
- Buildpacks: https://buildpacks.io/
  - Google Cloud buildpacks: https://cloud.google.com/docs/buildpacks/overview
  - Node.js: https://cloud.google.com/docs/buildpacks/nodejs
  - Python: https://cloud.google.com/docs/buildpacks/python
  - Java: https://cloud.google.com/docs/buildpacks/java
- Distroless images: https://github.com/GoogleContainerTools/distroless
  - Example build for Java: https://github.com/GoogleContainerTools/distroless/blob/main/examples/java/Dockerfile
  - Python: https://github.com/GoogleContainerTools/distroless/blob/main/examples/java/Dockerfile
  - Node.js: https://github.com/GoogleContainerTools/distroless/blob/main/examples/nodejs/Dockerfile

General K8S

  - Children's illustrated guide to Kubernetes: https://youtu.be/4ht22ReBjno
  - Text version: https://www.cncf.io/the-childrens-illustrated-guide-to-kubernetes/
- The Kubernetes Documentary:
  - https://youtu.be/BE77h7dmoQU
  - https://youtu.be/318elIq37PE
- What is Kubernetes: https://kubernetes.io/docs/concepts/overview/
- Nice "What is GKE" sketch: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/GKE.jpg

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engine/docs/concepts/cluster-architecture

- GKE editions (Standard vs Enterprise): https://cloud.google.com/kubernetes-engine/docs/concepts/gke-editions
- Google Kubernetes Engine docs: https://cloud.google.com/kubernetes-engine/docs/concepts/kubernetes-engine-overview
- Pod concepts: https://cloud.google.com/kubernetes-engine/docs/concepts/pod
- YAML cheat sheet: https://lzone.de/cheat-sheet/YAML
- **Kubernetes resource (kind/object) list: https://kubernetes.io/docs/reference/kubectl/overview/#resource-types
- **kubectl docs:
  - https://kubernetes.io/docs/reference/kubectl/quick-reference/
  - https://unofficial-kubernetes.readthedocs.io/en/latest/user-guide/kubectl-cheatsheet/
  - https://kubernetes.io/docs/reference/generated/kubectl/kubectl-commands
- Working with k8s yaml in VS Code / Cloud Shell: https://cloud.google.com/code/docs/vscode/yaml-editing
- Pods: https://cloud.google.com/kubernetes-engine/docs/concepts/pod
- Deployments: https://cloud.google.com/kubernetes-engine/docs/concepts/deployment
  - https://kubernetes.io/docs/concepts/workloads/controllers/deployment/
- StatefulSets: https://cloud.google.com/kubernetes-engine/docs/concepts/statefulset
- StatefulSet demo: https://kubernetes.io/docs/concepts/workloads/controllers/statefulset/
- Storage options: https://cloud.google.com/kubernetes-engine/docs/concepts/storage-overview
- Job/CronJob:
  - https://kubernetes.io/docs/concepts/workloads/controllers/job/
- Info on pod resource specifications:
  - https://kubernetes.io/docs/concepts/configuration/manage-resources-containers/
  - https://cloud.google.com/blog/products/gcp/kubernetes-best-practices-resource-requests-and-limits
- Pod overhead: https://kubernetes.io/docs/concepts/scheduling-eviction/pod-overhead/
- Controlling pod placement (affinity, taints):
  - https://kubernetes.io/docs/concepts/scheduling-

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engine/docs/how-to/horizontal-pod-autoscaling
- Kubernetes custom Application resource template (Marketplace requirement): https://github.com/kubernetes-sigs/application
- Helm package manager: https://helm.sh/
- Skaffold (another k8s installer): https://skaffold.dev/
- Kustomize: https://kustomize.io/
- Cool instance calculator: https://learnk8s.io/kubernetes-instance-calculator
- Create and manage multiple similar clusters: https://gardener.cloud/
- Cloud Deploy: https://cloud.google.com/deploy/docs/overview

GKE Networking

- GKE network overview: https://cloud.google.com/kubernetes-engine/docs/concepts/network-overview
- **GKE networking best practices: https://cloud.google.com/kubernetes-engine/docs/best-practices/networking
- Services: https://cloud.google.com/kubernetes-engine/docs/concepts/service
- LoadBalancer service options: https://cloud.google.com/kubernetes-engine/docs/concepts/service-load-balancer
- GKE Ingress: https://cloud.google.com/kubernetes-engine/docs/concepts/ingress
- Kubernetes ingress: https://kubernetes.io/docs/concepts/services-networking/ingress/
- GKE Gateways (a better way to do ingress): https://cloud.google.com/kubernetes-engine/docs/concepts/gateway-api
- NetworkPolicy documentation: https://kubernetes.io/docs/concepts/services-networking/network-policies/
- Container native load balancing: https://cloud.google.com/kubernetes-engine/docs/how-to/container-native-load-balancing
- Multi-cluster services:
    - Overview: https://cloud.google.com/kubernetes-engine/docs/concepts/multi-cluster-services
    - Configuring: https://cloud.google.com/kubernetes-engine/docs/how-to/multi-cluster-services
- Using Cloud DNS for GKE: https://cloud.google.com/kubernetes-engine/docs/how-to/cloud-dns

Some GKE specifics

- Vertical pod autoscaling: https://cloud.google.com/kubernetes-engine/docs/concepts/verticalpodautoscaler

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- GKE Cluster autoscaling: https://cloud.google.com/kubernetes-engine/docs/concepts/cluster-autoscaler
- Autoscaling using Cloud Monitoring metrics: https://cloud.google.com/kubernetes-engine/docs/tutorials/autoscaling-metrics
- Scale cluster up/down on schedule: https://cloud.google.com/kubernetes-engine/docs/tutorials/reducing-costs-by-scaling-down-gke-off-hours
- GKE docs on Windows Applications: https://cloud.google.com/kubernetes-engine/docs/how-to/deploying-windows-app
- Info on deploying Windows pods: https://cloud.google.com/blog/products/containers-kubernetes/how-to-deploy-a-windows-container-on-google-kubernetes-engine
- GKE Cost optimization: https://cloud.google.com/solutions/best-practices-for-running-cost-effective-kubernetes-applications-on-gke
- Building highly available clusters: https://cloud.google.com/blog/products/containers-kubernetes/best-practices-for-creating-a-highly-available-gke-cluster
- Multi-tenant clusters: https://cloud.google.com/kubernetes-engine/docs/best-practices/enterprise-multitenancy
- Multi-tenant logging: https://cloud.google.com/stackdriver/docs/solutions/gke/multi-tenant-logging
- Autopilot clusters: https://cloud.google.com/blog/products/containers-kubernetes/introducing-gke-autopilot
  - Autopilot documentation: https://cloud.google.com/kubernetes-engine/docs/concepts/autopilot-overview
  - Comparing GKE Standard and Autopilot: https://cloud.google.com/kubernetes-engine/docs/resources/autopilot-standard-feature-comparison
  - Setting resource requests and limits: https://cloud.google.com/kubernetes-engine/docs/concepts/autopilot-resource-requests
- GKE container image streaming:
  - https://cloud.google.com/blog/products/containers-kubernetes/introducing-container-image-streaming-in-gke
  - https://cloud.google.com/kubernetes-engine/docs/how-to/image-streaming
- GKE Backup: https://cloud.google.com/kubernetes-engine/docs/add-on/backup-for-gke/concepts/backup-for-gke

Monitoring/Troubleshooting

- Monitoring, logging, and debugging: https://kubernetes.io/docs/tasks/debug/
- **Operations suite for GKE: https://cloud.google.com/stackdriver/docs/solutions/gke/installing
- Customizing cluster logs:
  - https://cloud.google.com/community/tutorials/kubernetes-engine-customize-fluentbit
  - (legacy fluentd) https://cloud.google.com/architecture/customizing-stackdriver-logs-fluentd
- Troubleshooting Kubernetes: https://cloud.google.com/kubernetes-

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    - prometheus
    - Managed Service for Prometheus (GMP) Customer-Sharable Deck [public]
- Grafana: https://grafana.com/
- Open source Istio mesh visualization with Kiali: https://kiali.io/
- Configure Liveness, Readiness and Startup Probes: https://kubernetes.io/docs/tasks/configure-pod-container/configure-liveness-readiness-startup-probes/

GKE Security

- GKE security overview: https://cloud.google.com/kubernetes-engine/docs/concepts/security-overview
- GKE Hardening your cluster (and links to other security articles): https://cloud.google.com/kubernetes-engine/docs/how-to/hardening-your-cluster
- NSA Kubernetes Hardening guide: https://media.defense.gov/2022/Aug/29/2003066362/-1/-1/0/CTR_KUBERNETES_HARDENING_GUIDANCE_1.2_20
- Role Based Access Control (RBAC):
    - https://cloud.google.com/kubernetes-engine/docs/how-to/role-based-access-control
    - https://kubernetes.io/docs/reference/access-authn-authz/rbac/
- **Pod Security Standards
    - https://cloud.google.com/kubernetes-engine/docs/how-to/pod-security-policies-with-gatekeeper
    - https://kubernetes.io/docs/concepts/security/pod-security-standards/
    - Enforce Pod Security Standards with Namespace Labels: https://kubernetes.io/docs/tasks/configure-pod-container/enforce-standards-namespace-labels/
- Pod admission controllers: https://kubernetes.io/docs/concepts/security/pod-security-admission/
- Gatekeeper for pod policy management: https://cloud.google.com/kubernetes-engine/docs/how-to/pod-security-policies-with-gatekeeper
- GKE Private clusters: https://cloud.google.com/kubernetes-engine/docs/concepts/private-cluster-concept
- GKE related IAM roles: https://cloud.google.com/iam/docs/understanding-roles#kubernetes-engine-roles
- Binary Auth: https://cloud.google.com/binary-authorization/docs
- Workload Identity:
    - Concepts: https://cloud.google.com/kubernetes-engine/docs/concepts/workload-identity
    - Setting it up: https://cloud.google.com/kubernetes-engine/docs/how-to/workload-identity
- Google Groups for RBAC:
    - https://cloud.google.com/kubernetes-engine/docs/how-to/google-groups-rbac
    - https://cloud.google.com/kubernetes-engine/docs/how-to/google-groups-rbac

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Creating a Cluster

- Local dev K8S cluster
    - Minikube: https://minikube.sigs.k8s.io/docs/
    - Kind: https://kind.sigs.k8s.io/
        - Kink, a Kind helper: https://github.com/Trendyol/kink
- Installing on-prem with kOps: https://github.com/kubernetes/kops
- Other options for installing and setting up Kubernetes:
    - https://kubernetes.io/docs/setup/
    - Tools: https://kubernetes.io/docs/setup/production-environment/tools/
- Creating a GKE cluster in Google Cloud: https://cloud.google.com/kubernetes-engine/docs/how-to/creating-a-zonal-cluster

Custom Resources and Operators

- 3p operators: https://operatorhub.io/
- Custom Resources: https://kubernetes.io/docs/concepts/extend-kubernetes/api-extension/custom-resources/
- Example creating a CRD: https://developer.ibm.com/tutorials/kubernetes-custom-resource-definitions/
- Java example: https://medium.com/xgeeks/kubernetes-dummy-operator-in-java-6b2f26198a55
- Example creating an operator: https://developers.redhat.com/articles/2021/09/07/build-kubernetes-operator-six-steps
- Creating an operator in Java; https://github.com/java-operator-sdk/java-operator-sdk
- Operator pattern: https://kubernetes.io/docs/concepts/extend-kubernetes/operator/
- Installing a CRD: https://kubernetes.io/docs/tasks/extend-kubernetes/custom-resources/custom-resource-definitions/
- K8S client libraries: https://kubernetes.io/docs/reference/using-api/client-libraries/
- Operator SDK: https://sdk.operatorframework.io/

Istio

- Istio homepage: https://istio.io/
- About: https://istio.io/latest/about/service-mesh/
- **Istio architecture overview: https://istio.io/latest/docs/ops/deployment/architecture/
- Istio deployment models: https://istio.io/docs/ops/deployment/deployment-models/

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- Istio installation: https://istio.io/latest/docs/setup/install/
- Istio security:
  - https://istio.io/latest/docs/concepts/security/
  - https://istio.io/latest/docs/reference/config/security/
- Troubleshooting multicluster: https://istio.io/latest/docs/ops/diagnostic-tools/multicluster/
- Some Istio performance info: https://istio.io/docs/ops/deployment/performance-and-scalability/
- Istio traffic management:
  - ** https://istio.io/latest/docs/concepts/traffic-management/
  - https://istio.io/latest/docs/tasks/traffic-management/
- More than you ever wanted to know about mTLS (The Illustrated TLS Connection): https://tls12.xargs.org/
- Open source mesh visualization with Kiali: https://kiali.io/
- Circuit Breaking example: https://istio.io/latest/docs/tasks/traffic-management/circuit-breaking/

Anthos / GKE Enterprise

General

- Doc homepage: https://cloud.google.com/anthos/docs
- **Technical overview: https://cloud.google.com/anthos/docs/concepts/overview
- Reference architectures: https://cloud.google.com/anthos/docs/architecture/hybrid-environment
- Anthos overview sketch: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/Anthos.jpg
- **Deployment options (Features and where they work): https://cloud.google.com/anthos/deployment-options
- Anthos vs. competitors: https://competitiveinsightshub.withgoogle.com/area/21/matrix/1487
- How fleets work: https://cloud.google.com/anthos/fleet-management/docs/fleet-concepts
- Fleet management (start here): https://cloud.google.com/anthos/fleet-management/docs
- **Setting up: https://cloud.google.com/anthos/docs/setup/overview
- Connect agent: https://cloud.google.com/anthos/fleet-management/docs/connect-agent
- Migrate to Containers:
  - Getting started: https://cloud.google.com/migrate/anthos/docs/getting-started
  - Planning: https://cloud.google.com/migrate/containers/docs/planning-best-practices
  - Compatable OSs: https://cloud.google.com/migrate/containers/docs/compatible-os-versions
  - Architecture: https://cloud.google.com/migrate/anthos/docs/architecture
- Bank of Anthos demo app: https://github.com/GoogleCloudPlatform/bank-of-anthos
- GKE Identity Service: https://cloud.google.com/anthos/identity
- Knative serving (Cloud Run for Anthos): https://cloud.google.com/anthos/run/docs/setup

On-Prem / Other Clouds

- Setup (start here): https://cloud.google.com/anthos/docs/setup/on-premises

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cloud/aws/quickstart
    - Azure: https://cloud.google.com/anthos/clusters/docs/multi-cloud/azure/deploy-app
- Regional support:
    - AWS: https://cloud.google.com/anthos/clusters/docs/multi-cloud/aws/concepts/gcp-managed-regions
    - Azure: https://cloud.google.com/anthos/clusters/docs/multi-cloud/azure/reference/supported-regions
- Hardening on-prem (VMware) clusters: https://cloud.google.com/anthos/clusters/docs/on-prem/latest/how-to/hardening-your-cluster
- Manage Identity with GKE Identity Service: https://cloud.google.com/anthos/clusters/docs/on-prem/latest/concepts/authentication
- Load Balancer configuration options (on-prem VM Ware): https://cloud.google.com/anthos/clusters/docs/on-prem/latest/how-to/setup-load-balance

ABM / APM

- GKE Enterprise on Bare Metal overview: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/concepts/about-bare-metal
- ABM installation:
    - Prerequisite guides: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/installing/install-prereq
    - Plan your installation: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/installing/minimal-infrastructure
- Try ABM on GCE using Terraform: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/try/gce-vms-tf
- Network requirements: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/concepts/network-reqs
- Config file samples: https://cloud.google.com/anthos/clusters/docs/bare-metal/latest/reference/config-samples

Service Mesh / Cluster Networking

- Service Mesh documentation: https://cloud.google.com/service-mesh/docs
    - Overview: https://cloud.google.com/service-mesh/docs/overview
- Supported features:
    - Managed ASM: https://cloud.google.com/service-mesh/docs/managed/supported-features-mcp
    - In cluster: https://cloud.google.com/service-mesh/docs/supported-features
- ASM managed install guide: https://cloud.google.com/service-mesh/docs/install
- On-pre install: https://cloud.google.com/service-mesh/docs/unified-install/install-anthos-service-mesh
- Integrating IAP with Anthos Service Mesh: https://cloud.google.com/service-mesh/docs/unified-install/options/iap-integration
- About Multi-Cluster Services (MCS): https://cloud.google.com/kubernetes-engine/docs/concepts/multi-cluster-services
- Multi-cluster Anthos Ingress (to load balance over multiple clusters): https://cloud.google.com/kubernetes-engine/docs/concepts/multi-cluster-ingress

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- Configuring multi-cluster services:
  - Setting up in GKE: https://cloud.google.com/kubernetes-engine/docs/how-to/multi-cluster-services
  - Networking options: https://cloud.google.com/kubernetes-engine/docs/how-to/msc-setup-with-shared-vpc-networks
- Traffic Director: https://cloud.google.com/traffic-director/docs

Config Management

- Policy Controller, Config Sync, and Config Controller overview: https://cloud.google.com/anthos-config-management/docs/overview
- GitOps best practices: https://cloud.google.com/anthos-config-management/docs/concepts/gitops-best-practices
- Hierarchical (structured) Git repository: https://cloud.google.com/anthos-config-management/docs/concepts/hierarchical-repo
- Unstructured: https://cloud.google.com/anthos-config-management/docs/how-to/unstructured-repo
- Policy Controller
  - Overview: https://cloud.google.com/anthos-config-management/docs/concepts/policy-controller
  - Installation: https://cloud.google.com/anthos-config-management/docs/how-to/installing-policy-controller
  - Constraints: https://cloud.google.com/anthos-config-management/docs/latest/reference/constraint-template-library
- Hierarchy controller overview: https://cloud.google.com/anthos-config-management/docs/concepts/hierarchy-controller
- Anthos Config Connector overview: https://cloud.google.com/config-connector/docs/overview

Data:

- **GCP Storage Products: https://cloud.google.com/products/storage
- **Which storage should I use: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/storageoptions.jpg
- Designing optimal storage: https://cloud.google.com/architecture/storage-advisor
- **Database products: https://cloud.google.com/products/databases
- **Which database should I use: https://storage.googleapis.com/gweb-cloudblog-publish/images/Which-Database_v03-22-23.max-2000x2000.jpg
- **Classic data processing pipeline sketch: https://www.thecloudgirl.dev/data-analytics/data-analytics-pipeline
- Data center migration white paper: https://services.google.com/fh/files/misc/google_data_center_transformation.pdf
- End to end data and analytics demo: https://github.com/GoogleCloudPlatform/data-analytics-golden-demo
- Colossus:
  - Peak behind the scenes: https://cloud.google.com/blog/products/storage-data-transfer/a-peek-behind-colossus-googles-file-system

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BigQuery

- What is BigQuery: https://cloud.google.com/bigquery/docs/introduction
- Overview sketch: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/BigQuery.png
- bq command line reference: https://cloud.google.com/bigquery/docs/reference/bq-cli-reference
- Migrating from Redshift to BQ: https://cloud.google.com/bigquery/docs/migration/redshift-overview

Querying

- **BigQuery SQL reference: https://cloud.google.com/bigquery/docs/reference/standard-sql/query-syntax
- **SQL Function one page list: https://cloud.google.com/bigquery/docs/reference/standard-sql/functions-and-operators
- Querying nested repeated structures:
  - **Working with arrays and structures: https://cloud.google.com/bigquery/docs/arrays
  - Nice example using array_agg and structure to build a nested repeated column: https://medium.com/free-code-camp/exploring-a-powerful-sql-pattern-array-agg-struct-and-unnest-b7dcc6263e36
  - Overview: https://towardsdatascience.com/https-medium-com-martin-weitzmann-bigquery-sql-on-nested-data-cf9589c105f4
- Queries over external data:
  - External data sources: https://cloud.google.com/bigquery/external-data-sources
  - Federated queries: https://cloud.google.com/bigquery/docs/federated-queries-intro
  - **Query from Cloud Storage: https://cloud.google.com/bigquery/external-data-cloud-storage
- Scheduling queries: https://cloud.google.com/bigquery/docs/scheduling-queries
- Dry run query: https://cloud.google.com/bigquery/docs/dry-run-queries
- User Defined Functions (UDFs):
  - https://cloud.google.com/bigquery/docs/reference/standard-sql/user-defined-functions

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sql/remote-functions

- Multi-statement transactions (scripts): https://cloud.google.com/bigquery/docs/reference/standard-sql/transactions
- Multi-statement queries (scripts): https://cloud.google.com/bigquery/docs/multi-statement-queries
- Stored Procedures: https://cloud.google.com/bigquery/docs/procedures
- BQ Sessions: https://cloud.google.com/bigquery/docs/sessions-intro
- Example utils for BigQuery: https://github.com/GoogleCloudPlatform/bigquery-utils
- BigQuery ML:
  - **BigQuery ML Intro: https://cloud.google.com/bigquery/docs/bqml-introduction
  - **Good BigQuery logistic regression how to: https://cloud.google.com/bigquery/docs/logistic-regression-prediction

Performance, troubleshooting, and cost control

- **Intro to optimizing BigQuery performance: https://cloud.google.com/bigquery/docs/best-practices-performance-overview
- **Execution details info: https://cloud.google.com/bigquery/query-plan-explanation
- **Using the execution graph tab: https://cloud.google.com/bigquery/docs/query-insights
- Anti-pattern recognition tool: https://cloud.google.com/blog/products/data-analytics/bigquery-anti-pattern-recognition-tool-optimizes-performance/
- 14 ways to tune your SQL: https://towardsdatascience.com/14-ways-to-optimize-bigquery-sql-for-ferrari-speed-at-honda-cost-632ec705979#38ea
- Nice blog post on query optimization: https://cloud.google.com/blog/topics/developers-practitioners/bigquery-admin-reference-guide-query-optimization
- **Getting info about BigQuery with INFORMATION_SCHEMA: https://cloud.google.com/bigquery/docs/information-schema-intro
- Getting job metadata (INFORMATION_SCHEMA.JOBS*): https://cloud.google.com/bigquery/docs/information-schema-jobs
- Couple of cool examples using job information schema:
  - ** Estimate slot usage and cost for queries: https://cloud.google.com/bigquery/docs/information-schema-jobs#estimate_slot_usage_and_cost_for_queries

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using-information-schema

  - https://medium.com/google-cloud/bigquery-slot-squeezes-896d9e0f2fc
  - Git with the scripts: https://github.com/monishdoshi22/BQInfoSchema
- BQ Query Visualizer:
  - https://bqvisualiser.appspot.com/
  - https://github.com/GoogleCloudPlatform/professional-services/tree/main/tools/bq-visualizer
- Another query plan visualizer: https://github.com/shollyman/gcp-go-snippets/tree/main/queryplan_visualization
- How BigQuery caching works: https://cloud.google.com/bigquery/docs/cached-results
- Controlling costs: https://cloud.google.com/bigquery/docs/controlling-costs
- What is BI Engine: https://cloud.google.com/bigquery/docs/bi-engine-intro
- Error messages: https://cloud.google.com/bigquery/docs/error-messages
- BQ logs: https://cloud.google.com/bigquery/docs/reference/auditlogs
- BQ monitoring:
  - **In BQ: https://cloud.google.com/bigquery/docs/admin-resource-charts
  - Metrics for Metrics Explorer: https://cloud.google.com/monitoring/api/metrics_gcp#gcp-bigquery
- Edition pricing:
  - Overview: https://cloud.google.com/blog/products/data-analytics/introducing-new-bigquery-pricing-editions
  - Details: https://cloud.google.com/bigquery/docs/editions-intro
- Auto-scaling slots: https://cloud.google.com/bigquery/docs/slots-autoscaling-intro

Data ingestion and DML

- Nice blog post on ingesting data into BigQuery: https://cloud.google.com/blog/topics/developers-practitioners/bigquery-explained-data-ingestion
- Batch loading data: https://cloud.google.com/bigquery/docs/batch-loading-data
- BigQuery Transfer Service: https://cloud.google.com/bigquery-transfer/docs/introduction
- Streaming into BigQuery (legacy): https://cloud.google.com/bigquery/streaming-data-into-

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    - Streaming Java example: https://cloud.google.com/bigquery/docs/write-api-streaming
    - Batch load: https://cloud.google.com/bigquery/docs/write-api-batch
    - Best practices: https://cloud.google.com/bigquery/docs/write-api-best-practices

- Using DML: https://cloud.google.com/bigquery/docs/reference/standard-sql/data-manipulation-language
- DML and transactions: https://cloud.google.com/blog/products/data-analytics/dml-without-limits-now-in-bigquery

Schema and design

- BigQuery schemas: https://cloud.google.com/bigquery/docs/schemas
- Nested and repeated columns: https://cloud.google.com/bigquery/docs/nested-repeated
- **Views: https://cloud.google.com/bigquery/docs/views-intro
- **Materialized views: https://cloud.google.com/bigquery/docs/materialized-views-intro
- Cross region dataset replication: https://cloud.google.com/bigquery/docs/data-replication
- Defining and using table constraints for join optimization: https://cloud.google.com/blog/products/data-analytics/join-optimizations-with-bigquery-primary-and-foreign-keys
- Table functions: https://cloud.google.com/bigquery/docs/reference/standard-sql/table-functions
- Managing and updating table schemas: https://cloud.google.com/bigquery/docs/managing-table-schemas
- DDL: https://cloud.google.com/bigquery/docs/reference/standard-sql/data-definition-language
- Getting info about table metadata: https://cloud.google.com/bigquery/docs/information-schema-tables
- Table partitioning: https://cloud.google.com/bigquery/docs/partitioned-tables
- Managing partitioned tables: https://cloud.google.com/bigquery/docs/managing-partitioned-tables
- Table clustering: https://cloud.google.com/bigquery/docs/clustered-tables
- BigQuery quotas and limits: https://cloud.google.com/bigquery/quotas

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Security, backup, DLP

- BigQuery permissions and info on roles: https://cloud.google.com/bigquery/docs/access-control
- BigQuery table access control: https://cloud.google.com/bigquery/docs/control-access-to-resources-iam
- BigQuery column-level security: https://cloud.google.com/bigquery/docs/column-level-security
- Row-level security: https://cloud.google.com/bigquery/docs/row-level-security-intro
- Time travel (data at point in time, within last 7 days): https://cloud.google.com/bigquery/docs/time-travel
- Table snapshots (table backup): https://cloud.google.com/bigquery/docs/table-snapshots-intro
- DLP and BigQuery: https://cloud.google.com/dlp/docs/data-profiles

Dataform

- Overview: https://cloud.google.com/dataform/docs/overview
- Features: https://cloud.google.com/dataform/docs/features
- Dataform core
    - SQLx files: https://cloud.google.com/dataform/docs/dataform-core
    - Reference: https://cloud.google.com/dataform/docs/reference/dataform-core-reference
    - Samples: https://cloud.google.com/dataform/docs/reference/sample-scripts
    - Creating tables: https://cloud.google.com/dataform/docs/define-table
    - Incremntal tables: https://cloud.google.com/dataform/docs/incremental-tables
    - Assertions: https://cloud.google.com/dataform/docs/assertions
- Quickstart: https://cloud.google.com/dataform/docs/quickstart-create-workflow
- Intro to SQL workflows: https://cloud.google.com/dataform/docs/sql-workflows
- CLI: https://cloud.google.com/dataform/docs/use-dataform-cli
- Dataform Git repos: https://github.com/dataform-co
- Workflow settings: https://cloud.google.com/dataform/docs/configure-dataform

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- Using JavaScript in Dataform: https://cloud.google.com/dataform/docs/javascript-in-dataform
    - Includes: https://cloud.google.com/dataform/docs/reuse-code-includes
-

How BigQuery works

- Reliability in BQ: https://cloud.google.com/bigquery/docs/reliability-intro
- Disaster planning: https://cloud.google.com/bigquery/docs/reliability-disaster
- Big Query's native storage format (Capacitor):
    - https://cloud.google.com/blog/products/bigquery/inside-capacitor-bigquerys-next-generation-columnar-storage-format
    - https://cloud.google.com/blog/topics/developers-practitioners/bigquery-admin-reference-guide-storage
- BigQuery under the hood: https://cloud.google.com/blog/products/bigquery/bigquery-under-the-hood
- Query processing: https://cloud.google.com/blog/topics/developers-practitioners/bigquery-admin-reference-guide-query-processing
- How a query works (by a Spark guy): https://www.waitingforcode.com/data-engineering-gcp/gcp-bigquery-by-apache-spark-guy/read
- A little history on BigQuery: https://towardsdatascience.com/bigquery-the-unlikely-birth-of-a-cloud-juggernaut-b5ad476525b7
- Info on Slots: https://cloud.google.com/bigquery/docs/slots
- Shuffling data: https://cloud.google.com/blog/products/bigquery/in-memory-query-execution-in-google-bigquery

Bigtable:

- Big Table docs: https://cloud.google.com/bigtable/docs/
- Nice graphic: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/Bigtable.jpg
- **Big Table schema design: https://cloud.google.com/bigtable/docs/schema-design
- Understanding BigTable performance: https://cloud.google.com/bigtable/docs/performance
- Replication: https://cloud.google.com/bigtable/docs/replication-overview
- Application Profiles: https://cloud.google.com/bigtable/docs/app-profiles

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Cloud SQL, AlloyDB, and Spanner:

- Cloud SQL documentation: https://cloud.google.com/sql/docs
- Cloud SQL Maintenance: https://cloud.google.com/sql/docs/mysql/maintenance
- Cloud SQL Backups: https://cloud.google.com/sql/docs/mysql/backup-recovery/backups
- Using the Cloud SQL connector: https://cloud.google.com/sql/docs/mysql/connect-connectors
- Connecting to Cloud SQL from GKE: https://cloud.google.com/sql/docs/mysql/connect-kubernetes-engine
- Spanner docs: https://cloud.google.com/spanner/docs/concepts
- Cloud Girl Spanner: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/Spanner.jpg
- Spanner best practices: https://cloud.google.com/spanner/docs/best-practice-list
- Spanner Schema design and best practices: https://cloud.google.com/spanner/docs/schema-design
- Spanner white papers: https://cloud.google.com/spanner/docs/whitepapers
- Spanner emulator: https://cloud.google.com/spanner/docs/emulator
- Spanner compute capacity (Processing Units and Nodes): https://cloud.google.com/spanner/docs/compute-capacity
- AlloyDB: https://cloud.google.com/alloydb/docs/overview
    - Docs home: https://cloud.google.com/alloydb/docs
    - Create and connect to a DB: https://cloud.google.com/alloydb/docs/quickstart/create-and-connect#before-you-begin
    - Comparison to Cloud SQL and Spanner: https://gist.github.com/Weiyuan-Lane/8daeabab1371221a286f0131cab3065d

Cloud Storage:

- Google Cloud Storage: https://cloud.google.com/storage/docs/introduction
- Nice graphic: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/GCS.png
- Best practices: https://cloud.google.com/storage/docs/best-practices
- Using versioning: https://cloud.google.com/storage/docs/using-object-versioning

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- Access control: https://cloud.google.com/storage/docs/access-control
- Encryption options: https://cloud.google.com/storage/docs/encryption
- Cloud Transfer Service Docs: https://cloud.google.com/storage-transfer/docs
- On-prem transfer service: https://cloud.google.com/storage-transfer/docs/on-prem-overview
- CDN caching overview:
  - Setup: https://cloud.google.com/cdn/docs/using-cdn
  - Details: https://cloud.google.com/cdn/docs/caching
- Media CDN: CDN: https://cloud.google.com/cdn/docs
- gsutil docs (note: the new gcloud storage is way faster): https://cloud.google.com/storage/docs/gsutil
- Info on gcloud storage
  - https://cloud.google.com/blog/products/storage-data-transfer/new-gcloud-storage-cli-for-your-data-transfers
  - https://cloud.google.com/sdk/gcloud/reference/storage
- Host website out of cloud storage: https://cloud.google.com/storage/docs/hosting-static-website
- Autoclass: https://cloud.google.com/storage/docs/autoclass
- Managed folders (folder level permissions): https://cloud.google.com/storage/docs/managed-folders

Filestore

- Overview: https://cloud.google.com/filestore/docs/overview
- Quickstart: https://cloud.google.com/filestore/docs/quickstart-console
- How-to guides: https://cloud.google.com/filestore/docs/how-to

Firestore / Datastore:

- Firestore docs: https://cloud.google.com/firestore/docs
- Firestore best practices (latency): https://cloud.google.com/firestore/docs/best-practices
- Firestore graphic: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/firestore.jpg
- Firestore index types: https://firebase.google.com/docs/firestore/query-data/index-overview
- Firestore getting data: https://cloud.google.com/firestore/docs/query-data/get-data
- Firestore queries:

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https://cloud.google.com/datastore/docs/concepts/overview

- Datastore how to guides: https://cloud.google.com/datastore/docs/how-to
- Datastore queries: https://cloud.google.com/datastore/docs/concepts/queries
- Datastore transactions: https://cloud.google.com/datastore/docs/concepts/transactions
- GQL reference: https://cloud.google.com/datastore/docs/reference/gql_reference
- Firestore native vs. Datastore mode: https://cloud.google.com/datastore/docs/firestore-or-datastore

Pub/Sub

- Pub/Sub documentation home page: https://cloud.google.com/pubsub/docs
- Pub/Sub graphic: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/pubsub.jpg
- Subscriber info: https://cloud.google.com/pubsub/docs/subscriber
- Schemas: https://cloud.google.com/pubsub/docs/schemas
- Subscriber message filtering: https://cloud.google.com/pubsub/docs/filtering
- Pub/Sub lite: https://cloud.google.com/pubsub/lite/docs/how-to
- Pub/Sub std vs lite: https://cloud.google.com/pubsub/docs/choosing-pubsub-or-lite
- Exactly once messages: https://cloud.google.com/pubsub/docs/exactly-once-delivery
- Message ordering: https://cloud.google.com/pubsub/docs/ordering
- Local emulator: https://cloud.google.com/pubsub/docs/emulator
- Performance testing: https://cloud.google.com/blog/products/data-analytics/testing-cloud-pubsub-clients-to-maximize-streaming-performance
- Monitoring:
  - https://cloud.google.com/pubsub/docs/monitoring
  - https://cloud.google.com/monitoring/api/metrics_gcp#gcp-pubsub
- Message Replay: https://cloud.google.com/pubsub/docs/replay-overview

Operations/Observability:

- Operations suite graphic:

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https://github.com/GoogleCloudPlatform/microservices-demo

- Operations pricing: https://cloud.google.com/stackdriver/pricing
- Operations suite cost optimization: https://cloud.google.com/solutions/stackdriver-cost-optimization
- Ops unified Logging/Monitoring OpsAgent: https://cloud.google.com/stackdriver/docs/solutions/agents/ops-agent
    - Configuration options: https://cloud.google.com/stackdriver/docs/solutions/agents/ops-agent/configuration
- SRE
    - GCP Current status information: https://status.cloud.google.com/
    - Google SRE info: https://sre.google/
    - **Google's SRE books: https://sre.google/books/
    - The art of SLOs: https://landing.google.com/sre/resources/practicesandprocesses/art-of-slos/
    - SLI/SLO handout: https://cre.page.link/art-of-slos-handbook-pdf-letter
    - Starter SLI menu: https://sre.google/workbook/implementing-slos/#slis-for-different-types-of-services
    - High Availability: https://en.wikipedia.org/wiki/High_availability
    - Postmortem examples:
        - https://landing.google.com/sre/sre-book/chapters/postmortem/
        - https://www.atlassian.com/incident-management/postmortem/templates
        - https://response.pagerduty.com/after/post_mortem_template/
    - Disaster Recovery scenarios planning guide: https://cloud.google.com/solutions/dr-scenarios-planning-guide
- Monitoring
    - **GCP available metrics: https://cloud.google.com/monitoring/api/metrics
    - Metric kinds and types: https://cloud.google.com/monitoring/api/v3/kinds-and-types
    - **Using the Metrics Explorer: https://cloud.google.com/monitoring/charts/metrics-explorer
    - Monitoring documentation: https://cloud.google.com/monitoring/docs
    - Understanding Metric Scopes: https://cloud.google.com/monitoring/settings
    - Monarch, the time series DB:
        - https://medium.com/geekculture/understanding-monarch-googles-planet-scale-monitoring-system-60e59b63ac0c
        - https://research.google/pubs/pub50652/
    - How monitoring works: https://cloud.google.com/monitoring/api/v3/metrics

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    - Alerting behavior: https://cloud.google.com/monitoring/alerts/concepts-indepth
    - Log based alerts: https://cloud.google.com/logging/docs/alerting/log-based-alerts
    - SLO Monitoring: https://cloud.google.com/stackdriver/docs/solutions/slo-monitoring
    - Monitoring official resource names (monitored resources): https://cloud.google.com/monitoring/api/resources
    - Uptime checks: https://cloud.google.com/monitoring/uptime-checks
    - Synthetic monitoring:
        - Overview: https://cloud.google.com/monitoring/uptime-checks/introduction
        - Creating: https://cloud.google.com/monitoring/synthetic-monitors/create
        - Scanning for broken links: https://cloud.google.com/monitoring/synthetic-monitors/broken-links
    - Custom metrics: https://cloud.google.com/monitoring/custom-metrics
    - Monitoring Query Language: https://cloud.google.com/monitoring/mql
    - MQL example queries: https://cloud.google.com/monitoring/mql/examples
    - VM Monitoring Agent(legacy): https://cloud.google.com/monitoring/agent/install-agent
    - Ops unified Logging/Monitoring agent: https://cloud.google.com/stackdriver/docs/solutions/ops-agent
    - Managing agents across multiple VMs: https://cloud.google.com/monitoring/agent/monitoring/fleet-installation
    - Automating logging and monitoring agent installation with Agent Policies: https://cloud.google.com/stackdriver/docs/solutions/managing-agent-policies
    - Building custom dashboards: https://cloud.google.com/monitoring/charts/dashboards
    - Exporting metric data to BigQuery: https://cloud.google.com/architecture/stackdriver-monitoring-metric-export
    - Metric data retention: https://cloud.google.com/monitoring/quotas#data_retention_policy
- Logging
    - **Logging overview and info on available logs: https://cloud.google.com/logging/docs/overview#categories
    - **Cloud Audit Logs: https://cloud.google.com/logging/docs/audit
    - **Platform (product specific) logs: https://cloud.google.com/logging/docs/api/platform-logs

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    - Logging libraries: https://cloud.google.com/logging/docs/setup
    - Using Log Queries: https://cloud.google.com/logging/docs/view/building-queries
    - Log query language ref: https://cloud.google.com/logging/docs/view/logging-query-language
    - Log query examples: https://cloud.google.com/logging/docs/view/query-library-preview
    - Correlate log entries: https://cloud.google.com/logging/docs/view/correlate-logs
    - **LogEntry type docs: https://cloud.google.com/logging/docs/reference/v2/rest/v2/LogEntry
    - Custom indexing: https://cloud.google.com/logging/docs/analyze/custom-index
    - Network telemetry, VPC Flow logs:
        - Using: https://cloud.google.com/vpc/docs/using-flow-logs
        - Flow log overview and details: https://cloud.google.com/vpc/docs/flow-logs
    - VM Logging Agent: https://cloud.google.com/logging/docs/agent/installation
    - Customizing agent (fluentd): https://cloud.google.com/logging/docs/agent/configuration#configure
    - Load Balancer logging: https://cloud.google.com/load-balancing/docs/https/https-logging-monitoring
    - Storing logs in Cloud Storage: https://cloud.google.com/logging/docs/central-log-storage
    - Info on Access Transparency Logs: https://cloud.google.com/logging/docs/audit/access-transparency-overview
    - Log Router: https://cloud.google.com/logging/docs/routing/overview
    - Exporting log data to Splunk: https://cloud.google.com/solutions/exporting-stackdriver-logging-for-splunk
    - Aggregated sinks: https://cloud.google.com/logging/docs/export/aggregated_sinks
    - Managing logging and monitoring agent policies: https://cloud.google.com/stackdriver/docs/solutions/managing-agent-policies
    - Controlling log access with Log Views: https://cloud.google.com/logging/docs/logs-views
    - Log based alerts: https://cloud.google.com/logging/docs/alerting/log-based-alerts
- Error Reporting
    - Error Reporting: https://cloud.google.com/error-reporting/docs
    - Setting up: https://cloud.google.com/error-reporting/docs/setup

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- Trace
  - Trace: https://cloud.google.com/trace/docs
  - Trace setup: https://cloud.google.com/trace/docs/setup
  - Trace filters: https://cloud.google.com/trace/docs/trace-filters
- Profiler
  - Profiler: https://cloud.google.com/profiler/docs
  - About Profiler: https://cloud.google.com/profiler/docs/about-profiler
  - Setup: https://cloud.google.com/profiler/docs/about-profiler#profiling_agent
  - Reading Flame graphs: https://cloud.google.com/profiler/docs/concepts-flame
  - Filtering flame graphs: https://cloud.google.com/profiler/docs/filtering-profiles

Networking:

- Networking 101
  - CIDR explained: https://en.wikipedia.org/wiki/Classless_Inter-Domain_Routing
  - CIDR table: http://www.rjsmith.com/CIDR-Table.html
  - CIDR calculator: https://www.ipaddressguide.com/cidr
  - Info on Private networking (RFC 1918): https://en.wikipedia.org/wiki/Private_network
- Google Cloud VPCs
  - Networking (VPC) in GCP: https://cloud.google.com/vpc/docs/overview
  - GCP networking overview: https://cloud.google.com/blog/topics/developers-practitioners/google-cloud-networking-overview
  - Cloud Girl network graphic: https://storage.googleapis.com/gweb-cloudblog-publish/images/image3_oUMNvdq.max-2000x2000.jpg
  - IP Ranges for all GCP services: https://support.google.com/a/answer/10026322?hl=en
  - IP addresses: https://cloud.google.com/vpc/docs/ip-addresses
  - Valid subnet IP ranges: https://cloud.google.com/vpc/docs/subnets#valid-ranges
  - Peering VPC networks: https://cloud.google.com/vpc/docs/vpc-peering
  - Cloud DNS:

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    - Network bandwidth: https://cloud.google.com/compute/docs/network-bandwidth

- Network best practices
    - VPC design best practices and reference architectures: https://cloud.google.com/solutions/best-practices-vpc-design
    - Shared VPC Network docs: https://cloud.google.com/vpc/docs/shared-vpc
    - VPC limits and quotas: https://cloud.google.com/vpc/docs/quota
    - Hybrid connectivity options: https://cloud.google.com/hybrid-connectivity/
    - Interconnect providers: https://cloud.google.com/network-connectivity/docs/interconnect/concepts/service-providers
    - Serverless VPC access (put Cloud Run, App Engine, and Cloud Functions on VPC): https://cloud.google.com/vpc/docs/serverless-vpc-access
    - CDN caching overview: https://cloud.google.com/cdn/docs/caching
    - Media CDN: CDN: https://cloud.google.com/cdn/docs

- Network monitoring
    - VPC Flow Logs: https://cloud.google.com/vpc/docs/using-flow-logs
        - Details: https://cloud.google.com/vpc/docs/flow-logs
    - Network Intelligence Center: https://cloud.google.com/network-intelligence-center/docs
    - Packet Mirroring:
        - https://cloud.google.com/vpc/docs/packet-mirroring
        - Using (demo?): https://cloud.google.com/vpc/docs/using-packet-mirroring
    - Firewall logs: https://cloud.google.com/vpc/docs/firewall-rules-logging

- Load Balancing
    - **Choosing a Load Balancer: https://cloud.google.com/load-balancing/docs/choosing-load-balancer
    - Load Balancer sketch: https://github.com/priyankavergadia/GCPSketchnote/blob/main/images/CLB.jpg
    - Load Balancer logging: https://cloud.google.com/load-balancing/docs/https/https-logging-monitoring

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https://cloud.google.com/vpc/docs/firewalls

  - Hierarchical firewall policies: https://cloud.google.com/vpc/docs/firewall-policies

- VPC Service Controls:
  - Overview: https://cloud.google.com/vpc-service-controls/docs/overview
  - Creating a VPC Service Control access level: https://cloud.google.com/access-context-manager/docs/create-basic-access-level
  - Supported VPC Service control products https://cloud.google.com/vpc-service-controls/docs/supported-products
- Cloud VPN: https://cloud.google.com/network-connectivity/docs/vpn/concepts/overview
- Identity Aware Proxy (IAP) docs: https://cloud.google.com/iap/docs/concepts-overview
- Choosing a network connectivity product: https://cloud.google.com/network-connectivity/docs/how-to/choose-product
- TCP Forwarding through IAP (substitute for jump boxes): https://cloud.google.com/iap/docs/tcp-forwarding-overview
- Cloud Armor: https://cloud.google.com/armor/docs/cloud-armor-overview
- Private Service Connect: https://cloud.google.com/vpc/docs/private-service-connect

Deveopment/Terraform/Tools:

- Developing apps in Google Cloud: https://cloud.google.com/docs/application-development
- Nice list of Cloud Native tools: https://landscape.cncf.io/
- General Tools
  - Diagnostics for GCP: https://github.com/GoogleCloudPlatform/gcpdiag
  - Cool command line tricks: https://github.com/jlevy/the-art-of-command-line
  - vi cheat sheet http://www.atmos.albany.edu/daes/atmclasses/atm350/vi_cheat_sheet.pdf
  - sed cheat sheet: http://eriklievaart.com/cheat/linux/shell/sed.html
  - nano keyboard commands: https://staffwww.fullcoll.edu/sedwards/Nano/NanoKeyboardCommands.html
  - bash scripting cheat sheet: https://devhints.io/bash
  - RegEx:
    - https://regex101.com/

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- Git cheat sheet:
  - https://training.github.com/downloads/github-git-cheat-sheet.pdf
  - https://about.gitlab.com/images/press/git-cheat-sheet.pdf
- Git tutorial:
  - https://www.atlassian.com/git/tutorials/setting-up-a-repository
  - https://git-scm.com/docs/gittutorial
- Git remote/fetch/push/pull: https://www.atlassian.com/git/tutorials/syncing
- Git workflows: https://www.atlassian.com/git/tutorials/comparing-workflows
- Maven in 5 min: https://maven.apache.org/guides/getting-started/maven-in-five-minutes.html
- Markdown (md) cheatsheet:
  - https://www.markdownguide.org/cheat-sheet/
  - https://wordpress.com/support/markdown-quick-reference/
- Check your cron job scheduling spec: https://crontab.guru/
- Fake data creator: https://mockaroo.com/
- Dev/Ops and CI/CD
  - DevOps in Google Cloud: https://cloud.google.com/architecture/devops
  - CI/CD developers hub: https://cloud.google.com/docs/ci-cd
  - Cloud Build: https://cloud.google.com/cloud-build/docs
    - Cloud Build example: https://cloud.google.com/cloud-build/docs/quickstart-build
    - Creating basic cloudbuild.yaml: https://cloud.google.com/build/docs/configuring-builds/create-basic-configuration
    - Cloud Build builders: https://cloud.google.com/build/docs/cloud-builders
    - Community and custom builders: https://cloud.google.com/build/docs/configuring-builds/use-community-and-custom-builders
    - Private pools: https://cloud.google.com/build/docs/private-pools/private-pools-overview
    - Useful Medium articles: https://medium.com/tag/cloud-build

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        - Nice end to end example: https://cloud.google.com/binary-authorization/docs/getting-started-console
    - Container Registry: https://cloud.google.com/container-registry/docs/quickstart
    - Artifact Registry: https://cloud.google.com/artifact-registry/docs/quickstarts
    - Artifact Management: https://cloud.google.com/artifact-management/docs/overview
    - Setup gcloud for GitHub actions: https://github.com/google-github-actions/setup-gcloud
    - Google GitHub actions: https://github.com/google-github-actions
    - Example GitHub action workflows for GCP: https://github.com/google-github-actions/setup-gcloud#example-workflows
    - Cloud Deploy:
        - General info: https://cloud.google.com/deploy/docs/overview
        - Architecture: https://cloud.google.com/deploy/docs/architecture
        - Terminology: https://cloud.google.com/deploy/docs/terminology
        - For Cloud Run: https://cloud.google.com/deploy/docs/deploy-app-run
        - For GKE: https://cloud.google.com/deploy/docs/deploy-app-gke
        - For GKE Enterprise / Anthos: https://cloud.google.com/deploy/docs/anthos-targets
- Terraform
    - **Google Cloud provider documentation: https://registry.terraform.io/providers/hashicorp/google/latest/docs
    - **Google's blueprint Terraform helper modules: https://cloud.google.com/docs/terraform/blueprints/terraform-blueprints
    - Nice example for GCP: https://cloud.google.com/docs/terraform/get-started-with-terraform
    - Managing infrastructure as code with Terraform, Cloud Build, and GitOps : https://cloud.google.com/docs/terraform/resource-management/managing-infrastructure-as-code
    - Getting started with Terraform on GCP (tutorial): https://developer.hashicorp.com/terraform/tutorials/gcp-get-started
    - Terraform in Google Cloud: https://cloud.google.com/docs/terraform

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manager/docs/overview

        - Quickstart: https://cloud.google.com/infrastructure-manager/docs/deploy-vpc-with-terraform

    - Installing Terraform: https://developer.hashicorp.com/terraform/tutorials/gcp-get-started/install-cli
    - Reference:
    - Language reference: https://developer.hashicorp.com/terraform/language
    - Functions: https://www.terraform.io/language/functions
    - Style conventions: https://developer.hashicorp.com/terraform/language/syntax/style

- Secure foundations blueprint guide (how Google thinks you should build your org): https://services.google.com/fh/files/misc/google-cloud-security-foundations-guide.pdf
    - Example, best practice, foundation Terraform (based on the above guide): https://github.com/terraform-google-modules/terraform-example-foundation
- Refactoring: https://www.terraform.io/language/modules/develop/refactoring
- Store Terraform state file in GCS: https://cloud.google.com/docs/terraform/resource-management/store-state
- Other example Terraform scripts: https://github.com/terraform-google-modules
- Testing Terraform: https://www.hashicorp.com/blog/testing-hashicorp-terraform
- Managing IaC with Terraform, Cloud Build, and GitOps: https://cloud.google.com/architecture/managing-infrastructure-as-code
- Generate Terraform HCL docs from project/resource:
    - https://cloud.google.com/sdk/gcloud/reference/alpha/resource-config/bulk-export
    - Terraformer: https://github.com/GoogleCloudPlatform/terraformer
- Terraform version constraints: https://developer.hashicorp.com/terraform/language/expressions/version-constraints
- Terraform registry: https://registry.terraform.io/
- Google's Terraform policy validator (`gcloud terraform vet`): https://cloud.google.com/docs/terraform/policy-validation
    - Cloud Asset Inventory (CAI) asset types: https://cloud.google.com/asset-inventory/docs/supported-asset-types
    - GCP Policy library:

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https://cloud.google.com/deployment-manager/docs/dm-convert

Security:

- GCP Trusted Infrastructure: https://cloud.google.com/security/infrastructure
    - https://cloud.google.com/docs/security/infrastructure/design
    - Whitepaper on same: https://cloud.google.com/security/infrastructure/design/resources/google_infrastructure_whitepaper_fa.p
- Video on getting into a Google Data Center: https://youtu.be/kd33UVZhnAA
- OWASP top 10: https://owasp.org/www-project-top-ten/
- IAM
    - IAM basics: https://cloud.google.com/iam/docs/overview
    - **Understanding Google Cloud IAM Roles: https://cloud.google.com/iam/docs/understanding-roles
    - IAM conditions: https://cloud.google.com/iam/docs/conditions-overview
        - Common Expression Language (CLE) docs: https://github.com/google/cel-spec/blob/master/doc/langdef.md
    - Creating and managing Tags: https://cloud.google.com/resource-manager/docs/tags/tags-creating-and-managing
    - Tags and access control: https://cloud.google.com/iam/docs/tags-access-control
    - Full Resource Names: https://cloud.google.com/iam/docs/full-resource-names
        - https://cloud.google.com/asset-inventory/docs/resource-name-format
    - Deny policies: https://cloud.google.com/iam/docs/deny-access
    - Troubleshooting access: https://cloud.google.com/iam/docs/troubleshooting-access
    - Policy Analyzer: https://cloud.google.com/asset-inventory/docs/analyzing-iam-policy
    - Workload identity federation: https://cloud.google.com/iam/docs/workload-identity-federation
- Resource Management
    - Google Reference Architectures:https://cloud.google.com/architecture
    - Creating and managing organizational resources: https://cloud.google.com/resource-

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https://services.google.com/fh/files/misc/google-cloud-security-foundations-guide.pdf

- Google Cloud Architecture Framework guide: https://cloud.google.com/architecture/framework
- **Architecture diagram tool: https://googlecloudcheatsheet.withgoogle.com/architecture
- Creating and managing labels: https://cloud.google.com/resource-manager/docs/creating-managing-labels
- Products that support labels: https://cloud.google.com/resource-manager/docs/labels-supported-services
- Google Cloud setup checklist: https://cloud.google.com/docs/enterprise/setup-checklist
- Best practices for enterprises: https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations
- Best practices for planning accounts and organizations: https://cloud.google.com/architecture/identity/best-practices-for-planning
- Managing multiple organizations: https://cloud.google.com/resource-manager/docs/managing-multiple-orgs
- Essential Contacts: https://cloud.google.com/resource-manager/docs/managing-notification-contacts

- Service Accounts
  - Service Accounts: https://cloud.google.com/iam/docs/service-account-overview
  - Service Account keys: https://cloud.google.com/iam/docs/keys-create-delete
  - Creating short-lived Service Accounts: https://cloud.google.com/iam/docs/create-short-lived-credentials-direct

- Org policies
  - Intro to Org Policies: https://cloud.google.com/resource-manager/docs/organization-policy/overview
  - Understanding constraints: https://cloud.google.com/resource-manager/docs/organization-policy/understanding-constraints
  - Org policy constraints: https://cloud.google.com/resource-manager/docs/organization-policy/org-policy-constraints
  - Org policy by tagging: https://cloud.google.com/resource-manager/docs/organization-policy/tags-organization-policy
  - Restricting resource usage: https://cloud.google.com/resource-manager/docs/organization-policy/restricting-resources

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- Key mgmt. and Encryption
  - Secrets Manager:
    - https://cloud.google.com/secret-manager/docs/overview
    - https://cloud.google.com/secret-manager/docs/quickstart
  - Secret manager best practices: https://cloud.google.com/secret-manager/docs/best-practices
  - Google Cloud Key Management Service (KMS): https://cloud.google.com/kms/docs
  - KMS how-to's:https://cloud.google.com/kms/docs/how-tos
  - Info on key rotation: https://cloud.google.com/kms/docs/key-rotation
  - KMS IAM permissions and roles: https://cloud.google.com/kms/docs/reference/permissions-and-roles
  - Products supporting customer managed encryption keys (CMEK): https://cloud.google.com/kms/docs/using-other-products#cmek_integrations
  - Using customer managed encryption keys with External Key Manager (EKM): https://cloud.google.com/kms/docs/ekm
  - Information on using customer supplied encryption keys (CSEK): https://cloud.google.com/docs/security/encryption/customer-supplied-encryption-keys
  - Cloud HSM (Hardware Security Module): https://cloud.google.com/kms/docs/hsm
  - Encryption in transit: https://cloud.google.com/security/encryption-in-transit
  - Google Encryption at rest details:
    - https://cloud.google.com/security/encryption-at-rest/default-encryption

- Identity Provider (IdP)
  - Cloud Identity: https://cloud.google.com/identity/docs/overview
  - Best practices for planning accounts and organizations: https://cloud.google.com/architecture/identity/best-practices-for-planning
  - Best practices for federating Google Cloud with an external identity provider: https://cloud.google.com/architecture/identity/best-practices-for-federating
  - Identity reference architectures: https://cloud.google.com/architecture/identity/reference-architectures
  - Integrating Cloud Identity with 3rd-party user or IdPs: https://support.google.com/cloudidentity/topic/7558767

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    - Workforce Identity Federation: https://cloud.google.com/iam/docs/workforce-identity-federation
    - GCP Identity Platform: https://cloud.google.com/identity-platform/docs/how-to
    - Differences between Firebase and Identity Platform: https://cloud.google.com/identity-platform/docs/product-comparison
    - GCP Managed AD: https://cloud.google.com/managed-microsoft-ad/docs/quickstart-create-domain

- Other security tools and Best Practices
    - Security best practices center: https://cloud.google.com/security/best-practices
    - **Secure foundations blueprint guide: https://cloud.google.com/architecture/security-foundations/printable
    - Beyondcorp: https://cloud.google.com/beyondcorp
    - Compliance info: https://cloud.google.com/security/compliance
    - Web Security Scanner: https://cloud.google.com/security-command-center/docs/concepts-web-security-scanner-overview
    - Best practices for DDOS protection: https://cloud.google.com/files/GCPDDoSprotection-04122016.pdf
    - Security partner info: https://cloud.google.com/security/partners/
    - Security Command Center documentation:
        - https://cloud.google.com/security-command-center/docs
        - https://cloud.google.com/security-command-center/docs/concepts-security-command-center-overview
    - Forseti (security scanner) tutorials: https://cloud.google.com/community/tutorials/private-forseti-with-scc-integration
    - Forseti home page: https://forsetisecurity.org/
    - Five pillars of protection against ransomware attacks: https://cloud.google.com/blog/products/identity-security/5-pillars-of-protection-to-prevent-ransomware-attacks

Pricing/Cost

- **Price list for all resources: https://cloud.google.com/pricing/list
- **GCP Pricing Calculator: https://cloud.google.com/products/calculator

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GCP ACE Cert prep sheet Updated automatically every 5 minutes

Management: https://cloud.google.com/billing/docs/onboarding-checklist

- Committed use discounts: https://cloud.google.com/docs/cuds
- Billing best practices: https://cloud.google.com/docs/enterprise/best-practices-for-enterprise-organizations#billing_and_management
- **Cost optimization for developers and operators: https://cloud.google.com/architecture/cost-efficiency-on-google-cloud
- Operations suite cost optimization: https://cloud.google.com/solutions/stackdriver-cost-optimization
- Visualizing GCP billing using Looker: https://medium.com/google-cloud/visualize-gcp-billing-using-looker-99a90a462868
- Setting budgets and alerts: https://cloud.google.com/billing/docs/how-to/budgets
- Using BigQuery to analyze exported billing data: https://cloud.google.com/billing/docs/how-to/export-data-bigquery
- Lots of good Cloud Billing account how to's: https://cloud.google.com/billing/docs/how-to/account-management-overview
- "Cloud Computing Isn't as Cost Effective as Hoped": https://youtu.be/Vqgg-0fI4Co
- Was moving to cloud a mistake? https://www.infoworld.com/article/3676597/was-moving-to-cloud-a-mistake.html

Generative AI

- Foundational info and theory:
  - Introduction to Generative AI: https://www.youtube.com/watch?v=G2fqAlgmoPo
  - Rules of ML https://developers.google.com/machine-learning/guides/rules-of-ml
  - Large Language Models (LLMs) concepts guide: https://developers.generativeai.google/guide/concepts
  - Intro to LLMs: https://developers.google.com/machine-learning/resources/intro-llms
  - Understanding vector text embeddings: https://cloud.google.com/blog/topics/developers-practitioners/meet-ais-multitool-vector-embeddings
  - Understanding temperature: https://docs.cohere.com/docs/temperature
  - Understanding Top-k, Top-p: https://docs.cohere.com/docs/controlling-generation-with-top-k-top-p
  - Transformers, an illustrated guide: https://jalammar.github.io/illustrated-transformer/

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    - Visual view of Generative AI transformers: https://ig.ft.com/generative-ai/
    - **Prompt engineering techniques: https://www.promptingguide.ai/
    - Understanding ReAct: https://generativeai.pub/understand-react-and-how-it-works-in-three-minutes-f5f57a404a82
    - ROI of GenAI: https://services.google.com/fh/files/misc/the_roi_of_generative_ai.pdf
    - Google RAG Engine: https://cloud.google.com/vertex-ai/generative-ai/docs/rag-overview
- Overview of GenAI support in Vertex AI: https://cloud.google.com/vertex-ai/docs/generative-ai/learn/overview
- Available GenAI models and model parameters: https://cloud.google.com/vertex-ai/docs/generative-ai/learn/models
- Codey: https://cloud.google.com/vertex-ai/docs/generative-ai/code/code-models-overview
- Imagen:
    - Overview: https://cloud.google.com/vertex-ai/docs/generative-ai/image/overview
- Designing text prompts
    - Intro: https://cloud.google.com/vertex-ai/docs/generative-ai/learn/introduction-prompt-design
    - Overview: https://cloud.google.com/vertex-ai/docs/generative-ai/text/text-overview
- Model fine tuning, model distillation, and grounding:
    - Fine tuning: https://cloud.google.com/vertex-ai/docs/generative-ai/models/tune-models
    - Grounding: https://cloud.google.com/vertex-ai/generative-ai/docs/grounding/overview
    - Distilling purpose-built models: https://cloud.google.com/vertex-ai/docs/generative-ai/models/distill-text-models
    - Getting started with Vertex AI Search: https://cloud.google.com/generative-ai-app-builder/docs/try-enterprise-search
    - Getting started with Vertex AI Conversation: https://cloud.google.com/generative-ai-app-builder/docs/agent-data-store
    - Agents: https://cloud.google.com/dialogflow/vertex/docs
- Vertex AI Vector Search:
    - Overview: https://cloud.google.com/vertex-ai/docs/vector-search/overview
    - Quickstart: https://cloud.google.com/vertex-ai/docs/vector-search/quickstart

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- Gemini and Generative AI API coding and reference:
  - Python docs home page: https://cloud.google.com/python/docs/reference/aiplatform/latest/google.cloud.aiplatform
  - Using the VertexAI SDK: https://cloud.google.com/vertex-ai/docs/python-sdk/use-vertex-ai-python-sdk
  - And in its GitHub repo here: https://github.com/googleapis/python-aiplatform/
  - **Sample code: https://github.com/GoogleCloudPlatform/generative-ai
  - Overview of GenAI on Vertex: https://cloud.google.com/vertex-ai/docs/generative-ai/multimodal/overview
  - Getting started with Gemini: https://ai.google.dev/docs
  - Gemini API: https://cloud.google.com/vertex-ai/generative-ai/docs/model-reference/gemini
  - Try the API: https://cloud.google.com/vertex-ai/docs/generative-ai/start/quickstarts/quickstart-multimodal
  - Example code: https://ai.google.dev/examples?keywords=googleai
  - Function calling: https://cloud.google.com/vertex-ai/generative-ai/docs/multimodal/function-calling
  - Codelab on function calling: https://codelabs.developers.google.com/codelabs/how-to-cloud-run-gemini-function-calling?hl=en#0
  - Nice function calling example: https://glaforge.dev/posts/2023/12/22/gemini-function-calling/
- Notebook examples:
  - (GenAI): https://github.com/GoogleCloudPlatform/generative-ai/tree/main